1.1 Classical Mechanics Models

Many of the models used for studying molecules with large numbers of atoms are said to be classical because they rely on assumptions that allow the use of Newton's equations for the motion of the nuclei. Coming to this subject for the first time, one may wonder why we don't use quantum mechanics, since that is a more modern and comprehensive theory of the behavior of matter. A system of N atoms can be described in a probabilistic way using the Schrödinger equation. For illustration, consider a water molecule which has 10 electrons and 3 nuclei, each of which is identified with a point in three dimensions, thus the basic object describing the probability distribution for an isolated water molecule is a complex-valued function of 39 variables. Denote a point in this space by the coordinates $q_{1,x}, q_{1,y}, q_{1,z}, q_{2,x}, q_{2,y}, q_{2,z}, \ldots, q_{13,x}, q_{13,y}, q_{13,z}$. The Schrödinger equation itself is a partial differential equation of the following form:

$$i\hbar \frac{\partial \Phi}{\partial t} = -\hbar^2 \sum_{j=1}^{13} \frac{1}{2\mu_j} \left(\frac{\partial^2 \Phi}{\partial q_{j,x}^2} + \frac{\partial^2 \Phi}{\partial q_{j,y}^2} + \frac{\partial^2 \Phi}{\partial q_{j,z}^2} \right) + U_{\mathrm{P}}(q_{1,x}, q_{1,y}, \ldots, q_{13,z}) \Phi. \tag{1.1}$$

i is the square root of minus 1, $\hbar$ is Planck's constant, μ_j is the mass of the jth particle, Φ is the unknown "wave-function," and U_{P} is the primitive atomic potential energy function which includes formulas that model the repulsion of electrons due to their similar charge and the attraction of electrons to the positively charged nucleus of each atom. We cannot write down the solution of Schrödinger's equation in a simple form, so we would need eventually to solve the equation numerically, using a computer. Ultimately, the problem with the quantum mechanical approach is one of computational complexity. It is impossible (the "curse of dimensionality") to handle the numerical discretization of a partial differential equation in many variables, due to the rapid explosion in the number of grid points or basis functions that would be needed for approximation. Even the water molecule is beyond the reach of a full scale quantum mechanical treatment. In quantum mechanics we know the fundamental laws describing the system quite well, but the large number of independent spatial dimensions rules out any straightforward attack on the problem. In the words of Paul Dirac, "The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble" [106].

An assumption that is often made in order to reduce the complexity of the problem is that the relatively light electrons of the system move instantaneously, entirely slaved to the motion of the heavier nuclei (this is called the *Born-Oppenheimer approximation*). It is then found under certain circumstances that the nuclear coordinates can be viewed as the approximate solutions of a system of ordinary differential equations involving a Born-Oppenheimer potential energy

function $U = U_{\mathrm{BO}}$ depending only on their positions. The potential energy function U might be obtained by solving a time-independent version of Schrödinger's equation for just the electronic variables, but such *ab initio* methods can be very demanding (for the reasons mentioned above) and are only used in special cases. However, there is a lot of interest and great potential for *semi-classical* methods that vary the potential energy landscape ("surface-hopping" [378]) or model localized "wave packet" dynamics [240].

Using the Born-Oppenheimer approximation, it is possible to express the equations of motion of a system of nuclei, in some domain, by use of Newton's Second Law:

$$\text{mass} \times \text{acceleration} = \text{force},$$

where the forces are determined from the potential energy function U. Denoting the coordinates of the ith nucleus of an N-atom system by $q_{i,x}, q_{i,y}, q_{i,z}$, and the atomic mass by m_i, the equations of motion for the ith nucleus can be written out as

$$m_i \frac{\mathrm{d}^2 q_{i,x}}{\mathrm{d}t^2} = -\frac{\partial U}{\partial q_{i,x}}, \qquad m_i \frac{\mathrm{d}^2 q_{i,y}}{\mathrm{d}t^2} = -\frac{\partial U}{\partial q_{i,y}}, \qquad m_i \frac{\mathrm{d}^2 q_{i,z}}{\mathrm{d}t^2} = -\frac{\partial U}{\partial q_{i,z}}. \tag{1.2}$$

It is important to recognize that (1.2) does not, itself, give a complete description of the motion; it must be supplemented by initial conditions (positions and velocities given at some specified instant) for all atoms.

The potential U that appears here is much different than U_{P} that occurs in Schrödinger's equation (which governed the repulsive and attractive interactions of elementary particles). *Empirical force field methods* use assumptions about the form of the force field derived from first principles considerations for simplified systems. They represent a fairly dramatic simplification of the model. The pair interactions between isolated atoms may be well described by certain functional forms, but the empirical models go further in proposing that the potential energy U_{BO} can be replaced by an approximation constructed by summing up various contributions modelling the different interactions present; this approach is not quite consistent with the quantum mechanics of the system, but may allow sufficient accuracy for many practical purposes. A great deal of chemical insight, experimental data and simulation work goes into building the force fields used in molecular modelling, for examples see [9, 10, 53, 305, 394] and the references therein. U will be constructed from special functional forms to describe interactions of pairs, triples or quadruples of atomic nuclei, taking into account, at least to a limited extent, the interactions that would be predicted using quantum mechanics or which are observed in experiments. Some empirical potential models are described in the next section.

It is interesting that the passage from quantum mechanics to a classical model takes us from a linear partial differential equation (that of Schrödinger) to a nonlinear system of ordinary differential equations (those of Newton). Since linear problems are ones whose solutions (at least formally) can be written out, this might seem to be making the problem even more complex, or anyway just substituting

one kind of complexity for another, and indeed nonlinear equations are notoriously difficult to understand fully from a theoretical point of view. However, Newton's equations involve differentiating solutions only in time, and when we discretize the equations to put them on a computer we only have to introduce a grid in this one direction. The fact that there are many equations is not so serious, as the equations themselves are of a type that can easily be treated numerically. The total number of equations we need to solve only goes up linearly with N and the work involved in solving them numerically increases in a relatively modest "polynomial" way (i.e. as a polynomial function of N), compared to the exponential dependence of the computational work on the number of particles that is typically observed in the treatment of quantum mechanics. Even though there are simplified formulations of quantum mechanics and clever discretizations that make the complexity more manageable, the price is still much higher than that of the classical alternative.

The use of an empirical energy function limits the applicability of molecular dynamics, by ruling out certain behaviors that rely explicitly on chemical reactions; even when the assumption is appropriate, the derivation of an effective classical potential energy is nontrivial and may introduce modelling errors. The function U in (1.2) may thus be exceedingly complex in its structure, and the theoretical properties of the dynamical system are not easy to explain, yet, using Newton's equations, molecular simulations involving thousands or even millions of atoms become tractable. In this book we will work with the classical model since it has widespread significance and provides a very good starting point for numerical simulation. Most of the fundamental issues surrounding the theory of molecular dynamics and the construction of numerical methods are actually independent of the detailed features of the potential energy function—there is thus a remarkable generality to the molecular level models and methods used in chemistry, physics, engineering and biology.

Using a classical description based on a smooth, empirical force field is only one of several options available for molecular simulation. The simplest model for a molecular interaction potential is the hard-sphere model. We assume that each atom is an impenetrable sphere which interacts with other atoms via perfectly elastic collision with the atoms transferring, according to standard rules, momentum and energy to one another during the collisions. This was the first model treated in simulation [4] for understanding the statistical mechanics of liquids. Hard sphere simulations are sometimes referred to as "event-driven" since the steps in time are usually taken between the collision events (for recent examples, see [107]). We can think of hard sphere systems as being described by nonsmooth potentials ("infinite at short range"), and it is possible to model systems that include both hard core repulsion and smooth attractive potentials (we describe such methods in Sect. 3.7). Relying exclusively on hard spheres is very restrictive in terms of model accuracy and leads to less efficient propagation algorithms. Thus most modern simulations are based on smooth potentials.